1) Useful Results:

$$\sum_{i=0}^{L} p^i = \frac{1-p^{L+1}}{1-p} \xrightarrow{L\to\infty} \frac{1}{1-p}$$

$$\sum_{i=0,1}^{L} ip^i = \frac{p-(L+1)p^{L+1}+Lp^{L+2}}{(1-p)^2} \xrightarrow{L\to\infty} \frac{p}{(1-p)^2}$$

2) Physical Layer

Usually $P_{bit}(\gamma)$ is a monotonically decreasing function, so it is possible to do the following:

$$P[P_{bit} < P_{th}] = P[\gamma_{th} < \gamma] = P[P_{rx,th} < P_{rx}]$$

where the last form usually uses shadowing ($\psi \sim \mathcal{N}(\mu = 0, \sigma^2)$) which is gaussian. Recall that

$$P[\psi < a] = 1 - Q\left(\frac{a - \mu}{\sigma}\right)$$

Avg bit error:

$$\mathbb{E}[P_{bit}(\gamma)] = \int_0^\infty p(\gamma) P_{bit}(\gamma) d\gamma \to \int_0^\infty b e^{-a\gamma} d\gamma = \frac{b}{a}$$

recall

$$\int e^{-\alpha x} dx = -\frac{e^{-\alpha x}}{\alpha}$$

2-State Markov Model

$$\begin{aligned}
&\gamma_{th} = \gamma_0 \ln(2) \\
&P_{bit,G} = \frac{1}{2} P[\gamma_{th} < \gamma < \infty] = \frac{1}{2} \int_{\gamma_{th}}^\infty p(\gamma) P_{bit}(\gamma) d\gamma \\
&P_{bit,B} = \frac{1}{2} P[0 < \gamma < \gamma_{th}] = \frac{1}{2} \int_0^{\gamma_{th}} p(\gamma) P_{bit}(\gamma) d\gamma
\end{aligned}$$

MSE

Suppose to have a series of measurements of type $(d_i, M_i = P_{tx_i}/P_{rx_i})$ and a $P_L(d, \zeta)$ formula where ζ is a parameter to evaluate. The evaluation is done by minimizing the MSE, that is:

- Compute MSE:

$$MSE(\zeta) = \sum_i (P_L(d_i, \zeta) - M_i)^2$$

- Solve maximization problem:

$$\frac{d\, MSE(\zeta)}{d\zeta} = 0$$

3) Link Layer

$$P_{PDU} = 1 - (1 - P_{bit})^l$$
$$P_{LL} = 1 - \sum_{i=0}^{L-1} P_{PDU}^i (1 - P_{PDU}) = P_{PDU}^L$$
$$B_{LL} = \frac{\mathbb{E}[\text{bits succesfully delivered}]}{\mathbb{E}[\text{total time}]}$$
$$\mathbb{E}[t] = Lt_F P_{PDU}^L + \sum_i^L i t_F P_{PDU}^{i-1}(1 - P_{PDU}) = t_F \frac{1 - P_{PDU}^L}{1 - P_{PDU}}$$
$$\mathbb{E}[x] = (1 - P_{LL})l$$
$$S = \frac{\mathbb{E}[\text{All ok RXed PDUS}]}{\mathbb{E}[\text{All TXed PDUs}]}$$

ARQ Stuff

l = payload size
h = header size
R = rate

Packet Error Probability:

$$\boxed{P_{PDU} = 1 - (1 - P_{bit})^{l+h}}$$

Times:

- S&W:

$$t_G = t_{tx} + 2\tau_p + t_{ACK}$$

Attempts/Time for 1 success (L = # max allowed retx):

$$\boxed{\mathbb{E}[\text{attempts}] = \frac{1 - P_{PDU}^L}{1 - P_{PDU}} \xrightarrow{L\to\infty} \frac{1}{1 - P_{PDU}}}$$

- S&W

$$\mathbb{E}[t] = t_G \cdot \mathbb{E}[\text{attempts}] = t_G \frac{1 - P_{PDU}^L}{1 - P_{PDU}} \xrightarrow{L\to\infty} \frac{t_G}{1 - P_{PDU}}$$

- SR

$$\mathbb{E}[t] = t_{tx} \cdot \mathbb{E}[\text{attempts}] = t_{tx} \frac{1 - P_{PDU}^L}{1 - P_{PDU}} \xrightarrow{L\to\infty} \frac{t_{tx}}{1 - P_{PDU}}$$

LL Error rate:

$$\boxed{P_{LL} = P_{PDU}^L \xrightarrow{L\to\infty} 0}$$

Avg PDUs/bits rxed:

$$\boxed{\mathbb{E}[\text{rxed bits}] = (l + h)\mathbb{E}[\text{rxed PDUs}] = (l + h)(1 - P_{LL}) \xrightarrow{L\to\infty} l + h}$$

Throughput/Normalized Throughput:

$$\boxed{\rho = \frac{t_{tx}\mathbb{E}[\text{rxed PDUs}]}{\mathbb{E}[t]} \qquad S = \frac{\mathbb{E}[\text{rxed PDUs}]}{\mathbb{E}[\text{attempts}]} = \frac{(1 - P_{PDU})(1 - P_{LL})}{(1 - P_{PDU}^L)} \xrightarrow{L\to\infty} (1 - P_{PDU})}$$

- S&W

$$\rho = \frac{t_{tx}}{t_G} \frac{(1 - P_{PDU})(1 - P_{LL})}{(1 - P_{PDU}^L)} = \frac{t_{tx}}{t_G} S \xrightarrow{L\to\infty} \frac{t_{tx}}{t_G}(1 - P_{PDU})$$

- SR

$$\rho = S = \frac{(1-P_{PDU})(1-P_{LL})}{(1-P_{PDU}^L)} \overset{L\to\infty}{\longrightarrow} (1-P_{PDU})$$

Good put:

$$\boxed{\eta(l) = R\frac{l}{l+h}\rho}$$

- S&W:

$$\eta(l) = R\frac{l}{l+h}\frac{t_{tx}}{t_G}(1-P_{PDU}) \overset{L\to\infty}{\longrightarrow} R\frac{l}{l+h}\frac{t_{tx}}{t_G}(1-P_{PDU})$$

- SR

$$\eta(l) = R\frac{l}{l+h}(1-P_{PDU}) \overset{L\to\infty}{\longrightarrow} R\frac{l}{l+h}(1-P_{PDU})$$

useful derivatives:

$$\left[\frac{l}{l+h}\right]' = \frac{h}{(l+h)^2} \qquad \left[(1-P_{bit})^{l+h}\right]' = (1-P_{bit})^{l+h}\ln(1-P_{bit})$$

usually $\max l$ is found by solving:

$$\boxed{\ln(1-P_{bit})l^2 + \ln(1-P_{bit})hl + h = 0}$$

Counting

if IID, then

- $L \to \infty$ to count we use a gaussian rv such that:

$$\mathbb{E}[\text{count}] = \sum_{i=1}^{L} aip^{i-1}(1-p) = a\frac{1-(L+1)p^L + Lp^{L+1}}{1-p} \overset{L\longrightarrow\infty}{\to} a\frac{1}{1-p}$$

- $L < \infty$ to count we use final attempt + truncated gaussian

$$\mathbb{E}[\text{count}] = Lap^L + \sum_{i=1}^{L} aip^{i-1}(1-p) = a\frac{1-p^L}{1-p} \overset{L\longrightarrow\infty}{\to} a\frac{1}{1-p}$$

if not IID, then
Let:

- $P_s(i)$ be the success probability of the i-th event. usually it is $P_s(i) = P_i\prod_{k=1}^{i-1}(1-P_s(k))$
- The sum of all successes be $P_s = \sum_{i=1}^{L} P_s(i) \overset{L\to\infty}{\longrightarrow} 1$

$$\mathbb{E}[\text{count}] = La(1-P_s) + \sum_{i=1}^{L} iaP_s(i)$$

In a nested environment, a can be seen as the cost of the previous process. So for example we have 2 ARQ protocols with $\mathbb{E}[TX_1], \mathbb{E}[TX_2]$ computed individually. To find the total RETXs we must set $E[TX] = \mathbb{E}[TX_1]\mathbb{E}[TX_2]$

Error Correction

Given a FEC schema with (N, U) block codes, then

$$\boxed{P[\text{k errors}] = \binom{N}{k}P_{PDU}^k(1-P_{PDU})^{N-k}}$$

and let i be the last # of errors for correct decoding, then

$$P_{succ} = \sum_{k=0}^{i} P[\text{k errors}] \qquad P_{succ} = 1 - \sum_{k=i}^{K} P[\text{k errors}]$$

HARQ will have

$$S = \frac{U}{N} P_{succ} \qquad B = R \frac{l}{l+h} S$$

4) Transport Layer

In TCP some preliminary calculations are needed:

- From LL gather: $P_{PDU}, \mathbb{E}[\text{attempts}], \eta_{ARQ}(l)$
- If fragmentation is used, find: $SS, n^T_{PDU} = \lceil \frac{SS}{l} \rceil$
- Network Parameters:

$$P_{TCP} = 1 - \prod_i (1 - P_{TCP_i})$$
$$RTT = \sum_i RTT_i$$

Usually the next required formulas are provided, most importantly recall

$$B_{app} = \frac{8TCP_p}{SS} \cdot \min\{\eta_{ARQ}(l), B(RTT, P_{TCP})\}$$

Fragmentation ($TCP_l \geq PDU_l$)

Start from P_{PDU}

- find:

$$SS = IP_h + TCP_h + TCP_p$$
$$n^T_{PDU} = \left\lceil \frac{SS}{PDU_l} \right\rceil$$

- then

$$\begin{cases} P_{TCP_i} = 1 - (1 - P_{LL})^{n^T_{TCP}} \xrightarrow{L \to \infty} 0 \\ RTT_i = RTT_{\text{fixed}} \cdot \mathbb{E}[\text{attempts}] \end{cases}$$

Encapsulation ($TCP_l < PDU_l$)

Start from P_{PDU}

- find $P_{res} = P^L_{PDU}$
- find $P_{bit,res} = 1 - (1 - P_{res})^{\frac{1}{PDU_l}}$
- final probability:

$$\begin{cases} P_{TCP_i} = 1 - (1 - P_{bit,res})^{TCP_l} \xrightarrow{L \to \infty} 0 \\ RTT_i = RTT_{\text{fixed}} \end{cases}$$

TX Probability:

Let τ be the TX probability of n stations.
Probability at least one TX:

$$P_{tx} = P[\text{at least 1 tx}] = 1 - (1 - \tau)^n$$

$$P_s = P[\text{1 tx succesfull}] = \frac{P[\text{exactly 1 tx}]}{P_{tx}} = \frac{\tau(1 - \tau)^{n-1}}{P_{tx}}$$

The probability to have a collision is

$$P_c = P_{tx} - P_{tx}P_s = P_{tx}(1 - P_s)$$
$$P_{fail} = P^L_c$$

5) WI-Fi

Heavily relies on tables.

MCS (m_d/m_a)	Mod. Type	Mod. levels M	N_{BPSCS}	Coding Rate R	α	β
0	BPSK	2	1	1/2	0.1985	1.5090
1	QPSK	4	2	1/2	0.4460	1.6376
2	QPSK	4	2	3/4	0.1866	1.1715
3	16-QAM	16	4	1/2	0.1866	1.1715
4	16-QAM	16	4	3/4	0.5153	1.4078
5	64-QAM	64	6	2/3	0.4429	1.1660
6	64-QAM	64	6	3/4	0.3518	0.9434
7	64-QAM	64	6	5/6	0.2423	0.8956
8	256-QAM	256	8	3/4	0.6103	1.1777
9	256-QAM	256	8	5/6	0.3903	0.8170
10	1024-QAM	1024	10	3/4	0.6340	1.0321
11	1024-QAM	1024	10	5/6	0.3736	0.8442

Table I

Table I

Timing	value
T_{slot}	$9\ [\mu s]$
T_{sifs}	$16\ [\mu s]$
T_{difs}	$34\ [\mu s]$
CW_{min}	15 [slots]
CW_{max}	1023 [slots]
$T_{\text{data}}(\ell, m_d)$	$80 + \left\lceil \frac{36.75+\ell}{\text{BpS}(m_d)} \right\rceil T_{\text{symbol}}\ [\mu s]$
$T_{\text{ack}}(m_a)$	$80 + \left\lceil \frac{16.75}{\text{BpS}(m_a)} \right\rceil T_{\text{symbol}}\ [\mu s]$
BpS(m)	$N_{\text{SD}} N_{\text{BPSCS}} R/8$ [bytes/symbol]
$\eta_{\text{nominal}}(m_d)$	$\frac{\text{BpS}(m_d) \times 8}{T_{\text{symbol}}}$ [bit/s]

Table II

Table II

PHY	Modulation		R	N_{SS}	N_{SD}				T_{DFT}	T_{GI}		
	Name	N_{BPSCS}			20MHz	40MHz	80MHz	160MHz		Short	Medium	Long
802.11ax (HE)	BPSK	1	1/2	1 to 8	234	468	980	1960	12.8 µs	0.8 µs	1.6 µs	3.2 µs
	QPSK	2	1/2 & 3/4									
	16-QAM	4	1/2 & 3/4									
	64-QAM	6	1/2 & 2/3 & 3/4									
	256-QAM	8	2/3 & 5/6									
	1024-QAM	10	3/4 & 5/6									

Data Rates Table

Throughput:

$$\eta(q, l) = \frac{8l}{\zeta_1 + \dfrac{\zeta_2 q}{1 - 2q} + \dfrac{\zeta_3 q}{1 - q}}$$

with:

$$\begin{aligned}
\zeta_1 &= T_{\text{back}}(1) + T_{\text{difs}} + T_{\text{sifs}} + T_{\text{data}}(m_d, l) + T_{\text{ack}}(m_a) \\
&= T_{\text{back}}(1) + T_{\text{data}}(m_d, l) + T_{\text{wait}} \\
\zeta_2 &= W_{\text{min}} T_{\text{slot}} \\
\zeta_3 &= T_{\text{wait}} + T_{\text{data}}(m) - \frac{T_{\text{slot}}}{2}
\end{aligned}$$

$$\begin{aligned}
T_{\text{back}}(1) &= \frac{W_{\text{min}} - 1}{2} T_{\text{slot}} \\
T_{\text{wait}} &= T_{\text{difs}} + T_{\text{sifs}} + T_{\text{ack}}(m_a)
\end{aligned}$$

Remark.

As the theory suggests, the throughput is the ratio of rxed bits·txtime/total tx time. And thus the denominator is a good approximation for RTT

If no RETX is allowed, then the goodput is

$$G_{\text{no retx}} = \frac{(1-q)8l}{\zeta_1}$$